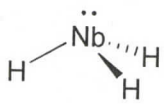
Inorganic / Materials Chemistry Cumulative Exam
1-25-19

Answer any five of the following ten questions (20 points each). Only your first five answers will be graded. Your answers are meant to provide evidence of broad, graduate-school level chemistry knowledge. Consider this in your choice of terminology and examples. Partial credit will be given.

Homogeneous Inorganic Chemistry

1. Propose a Molecular Orbital diagram for H_2O

2. What orbitals are of appropriate symmetry and energy to form the sigma bonds in NbH_3 (shown below)



3. Give three examples of kinetically stable metallocenes of the generic formula $(C_5H_5)_2M$. What criteria did you use to conclude they were "stable"

4. There are two common mechanisms of oxidative addition to a square planar d^8 metal species. Define these two mechanisms and give an example of each.

5. Define a weak field ligand and a strong field ligand. What properties / characteristics can be used to predict if a ligand is a strong field ligand or a weak field ligand.

Solid State and Materials Chemistry

6. What is the difference between the melting point of a polymer and the glass transition temperature of a polymer?

7. The three-dimensional extended coordination polymer Prussian Blue has the generic formula Fe_7CN_{18} and is insoluble in water. Prussian blue is prepared by mixing aqueous solutions of $Fe(ClO_4)_2$ (a green solid and solution) and $K_3Fe(CN)_6$ (a red solid and solution) which upon mixing immediately form an intense blue suspension and precipitate. What is the source of the intense blue color of Prussian Blue?

8. Two common electrode materials for routine cyclic voltammetry are HOPG (Highly Oriented Pyrolytic Graphite) and Glassy Carbon. Compare and contrast these two materials.

9. Compare and contrast a ferromagnetic solid, an antiferromagnetic solid and a diamagnetic solid. What properties are the same? What properties are different?

10. Compare and Contrast metallic bonding in solids and network covalent bonding in solids.

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$$\# reps = \frac{1}{h}\sum(g * \chi_r * \chi_i)$$

h = total operations in a group

g = number of operations in a class

χ_r = the character of the reducible representation

χ_i= the character of the irreducible representation

The transformation matrix for a C_n rotation around the z axis is

$$\begin{bmatrix} cos\,\theta & -sin\theta & 0 \\ sin\theta & cos\,\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$cos\frac{2\pi}{1} = 1, sin\frac{2\pi}{1} = 0$

$cos\frac{2\pi}{2} = -1, sin\frac{2\pi}{2} = 0$

$cos\frac{2\pi}{3} = -0.5, sin\frac{2\pi}{3} = \frac{\sqrt{3}}{2}$

$cos\frac{2\pi}{4} = 0, sin\frac{2\pi}{4} = 1$

$cos\frac{2\pi}{5} = \frac{1}{4}(-1+\sqrt{5}), sin\frac{2\pi}{5} = \frac{1}{4}\sqrt{10+2\sqrt{5}}$

$cos\frac{2\pi}{6} = 0.5, sin\frac{2\pi}{6} = \frac{\sqrt{3}}{2}$

Character Tables

C_2	E	C_2		
A	1	1	z, R_z	x^2, y^2, z^2, xy
B	1	-1	x, y, R_x, R_y	yz, xz

C_3	E	C_3	$(C_3)^2$		
A	1	1	1	z, R_z	x^2+y^2, z^2
E	1 1	e e^*	e^* e	x+iy; R_x+iR_y x-iy; R_x-iR_y	(x^2-y^2, xy) (yz, xz)

e = exp(2πi/3)

C_4	E	C_4	C_2	$(C_4)^3$		
A	1	1	1	1	z, Rz	x^2+y^2, z^2
B	1	-1	1	-1		x^2-y^2, xy
E	1 1	i -i	-1 -1	-i i	x+iy; R_x+iR_y x-iy; R_x-iR_y	(yz, xz)

C_{2v}	E	C_2	$\sigma_v(xz)$	$\sigma_v(yz)$		
A_1	1	1	1	1	z	x^2, y^2, z^2
A_2	1	1	-1	-1	R_z	xy
B_1	1	-1	1	-1	x, R_y	xz
B_2	1	-1	-1	1	y, R_x	yz

C_{3v}	E	$2C_3$	$3\sigma_v$		
A_1	1	1	1	z	x^2+y^2, z^2
A_2	1	1	-1	R_z	
E	2	-1	0	(x, y) (R_x, R_y)	(x^2-y^2, xy) (xz, yz)

C_{4v}	E	$2C_4$	C_2	$2\sigma_v$	$2\sigma_d$		
A_1	1	1	1	1	1	z	x^2+y^2, z^2
A_2	1	1	1	-1	-1	R_z	
B_1	1	-1	1	1	-1		x^2-y^2
B_2	1	-1	1	-1	1		xy
E	2	0	-2	0	0	(x, y) (R_x, R_y)	(xz, yz)

C_{6v}	E	$2C_6$	$2C_3$	C_2	$3\sigma_v$	$3\sigma_d$		
A_1	1	1	1	1	1	1	z	x2+y2, z2
A_2	1	1	1	1	-1	-1	Rz	
B_1	1	-1	1	-1	1	-1		
B_2	1	-1	1	-1	-1	1		
E_1	2	1	-1	-2	0	0	(x, y) (Rx, Ry)	(xz, yz)
E_2	2	=1	=1	2	0	0		(x^2-y^2, xy)

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C_{5v}	E	$2C_5$	$2(C_5)^2$	$5\sigma_v$		
A_1	1	1	1	1	z	x2+y2, z2
A_2	1	1	1	-1	Rz	
E_1	2	2cos(2π/5)	2cos(4π/5)	0	(x, y) (Rx, Ry)	(xz, yz)
E_2	2	2cos(4π/5)	2cos(2π/5)	0		(x2-y2, xy)

D_{2h}	E	C_2 (z)	C_2 (y)	C_2 (x)	i	σ (xy)	σ (xz)	σ (yz)		
A_g	1	1	1	1	1	1	1	1		x^2, y^2, z^2
B_{1g}	1	1	-1	-1	1	1	-1	-1	R_z	xy
B_{2g}	1	-1	1	-1	1	-1	1	-1	R_y	xz
B_{3g}	1	-1	-1	1	1	-1	-1	1	R_x	yz
A_u	1	1	1	1	-1	-1	-1	-1		
B_{1u}	1	1	-1	-1	-1	-1	1	1	z	
B_{2u}	1	-1	1	-1	-1	1	-1	1	y	
B_{3u}	1	-1	-1	1	-1	1	1	-1	x	

D_{3h}	E	$2C_3$	$3C'_2$	σ_h	$2S_3$	$3\sigma_v$		
A'_1	1	1	1	1	1	1		x^2+y^2, z^2
A'_2	1	1	-1	1	1	-1	R_z	
E'	2	-1	0	2	-1	0	(x, y)	(x^2-y^2, xy)
A''_1	1	1	1	-1	-1	-1		
A''_2	1	1	-1	-1	-1	1	z	
E''	2	-1	0	-2	1	0	(R_x, R_y)	(xz, yz)

D_{4h}	E	$2C_4$	C_2	$2C'_2$	$2C''_2$	i	$2S_4$	σ_h	$2\sigma_v$	$2\sigma_d$		
A_{1g}	1	1	1	1	1	1	1	1	1	1		x^2+y^2, z^2
A_{2g}	1	1	1	-1	-1	1	1	1	-1	-1	Rz	
B_{1g}	1	-1	1	1	-1	1	-1	1	1	-1		x^2-y^2
B_{2g}	1	-1	1	-1	1	1	-1	1	-1	1		xy
E_g	2	0	-2	0	0	2	0	-2	0	0	(Rx, Ry)	(xz, yz)
A_{1u}	1	1	1	1	1	-1	-1	-1	-1	-1		
A_{2u}	1	1	1	-1	-1	-1	-1	-1	1	1	z	
B_{1u}	1	-1	1	1	-1	-1	1	-1	-1	1		
B_{2u}	1	-1	1	-1	1	-1	1	-1	1	-1		
E_u	2	0	-2	0	0	-2	0	2	0	0	(x, y)	

T_d	E	$8C_3$	$3C_2$	$6S_4$	$6\sigma_d$		
A_1	1	1	1	1	1		$x^2+y^2+z^2$
A_2	1	1	1	-1	-1		
E	2	-1	2	0	0		$(2z^2-x^2-y^2, x^2-y^2)$
T_1	3	0	-1	1	-1	(Rx, Ry, Rz)	
T_2	3	0	-1	-1	1	(x, y, z)	(xy, xz, yz)